

FlowState auto-report, regenerated: e13_i94_rbc (scenarios/mndot_i94_wb_stpaul_weave_dc_cal_w1b_w2_rbc.yaml)

Generated: 2026-10-08T17:16:25Z by scripts/regenerate_report.py (stage p22_reports, code f0551e7a873c4a84bf22cd3eb79a1b48a7d91b8a).

Provenance

figures from replicate 6914975401685141156; tables from artifacts/validation_mndot_i94_wb_stpaul_weave_xlsfg_dc_cal_w1b_w2_rbc_gated.json

The figure replicate reproduces the battery's replicate 6914975401685141156: insertion, n_collisions, weave_releases are identical to the battery's record of that seed.

Item	Value
battery artifact	artifacts/validation_mndot_i94_wb_stpaul_weave_xlsfg_dc_cal_w1b_w2_rbc_gated.json, sha256 951dc6b75443
battery written	2026-10-08T05:42:10Z
scenario	scenarios/mndot_i94_wb_stpaul_weave_dc_cal_w1b_w2_rbc.yaml, sha256 d9eae2e7b164
replicates (battery)	n = 20 distinct seeds
criteria profile	fhwa_tat3_2004 – FHWA Traffic Analysis Toolbox Vol. III (2004, FHWA-HRT-04-040), §5.6 'Calibration Targets', Wisconsin DOT freeway model calibration criteria (captioned 'Table 4' on the HTML edition at https://ops.fhwa.dot.gov/trafficanalysistools/tat_vol3/sect5.htm, verified 2026-09-03; attributed there to WisDOT District 2, Paramics Calibration and Validation Guidelines, Technical Report I-33, June 2002). Transcribed row: 'GEH Statistic < 5 for Individual Link Flows: > 85% of cases' – strict '> 85%' (geh_pass_inclusive=False). Rows of that table NOT evaluated by FlowState: individual link flows within 15% / 100 veh/h / 400 veh/h by flow band for > 85% of cases; sum of all link flows within 5%; GEH < 4 for the sum of all link flows; journey times within 15% (or 1 min, if higher) for > 85% of cases; visual audits of speed-flow relationships and bottleneck queuing 'to analyst's satisfaction'. The table defines no RMSPE speed bound, so this profile produces no speeds_rmspe row (rmspe_max=None). The 2004 volume gives no fixed minimum run count (its §5 example uses ten replications to estimate a parameter within 5%), so min_seeds keeps FlowState's 20 (CLAUDE.md §0.6). The segment-speed RMSPE <= 15% bound is FlowState's own convention (CLAUDE.md §7.1, 'common microsim practice'); the 14-22 km/h emergent backward wave-speed band is the empirical stop-and-go literature value (flowstate_core.constants.WAVE_SPEED_BAND_KMH); the ring emergence/dampening rows reproduce Sugiyama et al. (2008) and Stern et al. (2018) per CLAUDE.md §3.2.1; min_seeds = 20 and the penetration {1,2,5,10,15,20}% x compliance {25,50,80,100}% grid are CLAUDE.md §0.6/§7.1 internal standards; so is the no_collisions model-integrity row (zero SUM0 collisions in every run, each run recording the counter; CLAUDE.md §3.3 and the owner decision of 2026-10-04) and the no_locks row (no run locked: no queue standing without discharge for the validation.locks duration; docs/I94_COLLAPSE_DIAGNOSIS.md, 2026-10-07), both present in every profile. None of these is prescribed by the cited DOT/FHWA documents. The wave-speed row is measured with the profile's wave_detector (validation.waves.WAVE_DETECTORS; default 'stack', chosen on the planted-stripe benchmark in validation.waves), and the evaluated row names that detector and its parameters.
config hash, battery	ad158ff561b1 (policy v4)
config hash, figure replicate	ad158ff561b1 (policy v4)

Item	Value
figure replicate	runs/p22/e13_i94_rbc/ad158ff561b1/6914975401685141156; seed 6914975401685141156; wall time 978.1 s
the battery's own auto-report	docs/reports/mndot_i94_wb_stpaul_weave_xlsfg_dc_cal_w1b_w2_rbc/report.m d (written on its VM from every replicate's metadata)

Seeds of the battery: 6914975401685141156, 134183728835869882, 2378473973028931053, 3747978530954135749, 677105600768189526, 661281422688282993, 3011106312394044631, 6953598295321596746, 165503670820534583, 6134032994440706937, 8557154790156791364, 6538422657834023852, 6904272788004776631, 6143473282319009404, 3944094060050347669, 4910985839736976611, 7382187975121682178, 5690692725577505498, 887972120279483394, 8026499204807041784.

What comes from where

Part of this report	Source
acceptance criteria, metric intervals, model integrity, observed data, notes, baseline gate	artifacts/validation_mndot_i94_wb_stpaul_weave_xlsfg_dc_cal_w1b_w2_rbc_gated.json: all 20 replicates of the battery
speed-contour figures	replicate 6914975401685141156 re-run with its trajectories by stage p22_reports (code f0551e7a873c4a84bf22cd3eb79a1b48a7d91b8a)
calibration artifacts	the figure replicate's metadata (the configuration fixes them)

Package versions (battery)

- eclipse-sumo: 1.27.1
- flowstate_core: 2.0.0
- libsumo: 1.27.1
- microsim: 2.0.0
- numpy: 2.5.2
- pandas: 3.0.5
- pyarrow: 25.0.1
- python: 3.12.15

Calibration artifacts used

Artifact	data_hash
artifacts/idm_i24_capacity_amax_k1.0.json	aa97dd93d2bf250ea23c3624f91afe8d7035a672fd13efbd53013709e12c7d56

Read from the figure replicate's metadata: the configuration, the same as the battery's, fixes them.

Observed data

The link-flow and segment-speed criteria are scored against this observed artifact: hourly volumes formed from the fully observed windows of each hour at every mainline station, and mean speeds per station segment (each station owns the span to the midpoints with its neighbours). Simulation time zero is the artifact's local start time; the run's warm-up, and any window the run does not cover to its end, are excluded, and a station-window the detector did not measure is skipped, never imputed — the coverage rows say how much of the grid was compared. A station whose cross-section lies outside the simulated position span is excluded from both comparisons rather than scored against a simulated flow of zero; the rows say how many were. When the artifact carries a detector-estimated backward wave speed, it is printed here as context for the band the simulated wave speed is scored against — it is a property of the corridor, not a target, and no criterion is evaluated from it.

Field	Value
artifact	artifacts/pl_rehearsal_2026-10-04/observations_calibration.json
corridor	I-94 WB
source provider	MnDOT RTMC Mayfly API
source dates	20260902, 20260903, 20260908, 20260915, 20260916
source url	https://data.dot.state.mn.us/mayfly
aggregation	mean over dates per window (weekday typical profile)
local clock time of simulation t = 0	05:30
observation window [s]	300
mainline stations compared	14
observation windows	48
windows inside the measurement window	42
station-windows with an observed flow [fraction]	1
station-windows with an observed speed [fraction]	1
link-hour comparisons (pooled over replicates)	840
speed cells compared (pooled over replicates)	11760
replicates scored against the observations	20
stations excluded (outside the simulated span)	0
detector-estimated backward wave speed (context, not a criterion)	median 19.1 km/h (IQR 18–23.9) from 7 of 13 station pairs; leave-one-date-out 18.5–20.8 km/h over 5 dates (fewest 5 pairs); the model's band is 14–22 km/h

Model integrity

What the battery artifact records about the simulation itself misbehaving, pooled over its replicates as the battery read them from every run's metadata. A collision or a lock is a model defect, not a traffic outcome; a count the battery did not record is reported as not recorded, never as zero.

- Collisions: 0 over 20 run(s) that record the counter; per run 0 [0, 0] (two-sided 95 % t-interval, n = 20); 0 per 1,000 departed vehicles (0 over 608243 departed in 20 run(s), whole runs including warm-up).
- Locks: 0 of 20 run(s) locked (0 %; 95 % Clopper-Pearson interval 0–16.8 %). A lock is a queue standing with no discharge past a point for at least 10 min.
- Insertion: 621380 vehicles planned over 20 run(s), 608243 departed (0.979 of plan on average, lowest 0.971), 28415.7 arrived per run (over 20 of 20); verdict: backlog: 2 % of planned vehicles never departed.
- Weave exits at on-ramp 745524613 (exit C-D split 18208090): 436 of 29359 reached exiters (1.49 %) were given up at the gore's end and rerouted through over 20 run(s), within the 2 % threshold.
- Weave exits at on-ramp 769818012 (exit off-ramp 18207598): 976 of 77449 reached exiters (1.26 %) were given up at the gore's end and rerouted through over 20 run(s), within the 2 % threshold.
- Weave rules at on-ramp 745524613 (exit C-D split 18208090): W1b released 281 of 20340 entrance departures (1.38 %) over 20 run(s), above the 1 % design bound; W2 (20 of 20 run(s) with all three guards on): handback skips 3666, close-leader withholds 681, opposing deferrals 7018, of them vetoes 6946.
- Weave rules at on-ramp 769818012 (exit off-ramp 18207598): W1b released 6 of 91997 entrance departures (0.00652 %) over 20 run(s), within the 1 % design bound; W2 (20 of 20 run(s) with all three guards on): handback skips 2924, close-leader withholds 93, opposing deferrals 16372, of them vetoes 15589.

Waiting measures

Each replicate's travel time and total delay including the time spent waiting on ramps and to enter the

network, with replicate intervals as above.

Measure	Mean	Lower	Upper	n	Underpowered
insertion_delay_veh_h	1097	1035	1158	20	no
meter_wait_veh_h	0	0	0	20	no
total_delay_incl_waiting_veh_h	5461	5325	5598	20	no
mean_tt_incl_waiting_s	918.9	901.6	936.3	20	no
p90_tt_incl_waiting_s	2004	1965	2043	20	no
n_censored	2653	2589	2717	20	no
n_demand_veh	2.82e+04	2.82e+04	2.82e+04	20	no
n_not_inserted	656.9	603.4	710.3	20	no
n_tt_incl_waiting_veh	2.777e+04	2.776e+04	2.777e+04	20	no
n_tt_censored	2220	2155	2284	20	no

Baseline gate

Baseline gate FAILED (docs/FRISCO_PROTOCOL.md section 6): C1 link flows (calibration), C3 speeds (calibration), C1 link flows (validation), C3 speeds (validation), C4 wave speed (calibration). No strategy recommendation may be made from this model.

Check	Day set	Status	Gating	Statement
replicates	all runs	PASS	yes	Replicates: 20 seeded replicate(s); the protocol scores every check over at least 20.
days	both day sets	PASS	yes	Day sets: the calibration-day artifact holds exactly the split's 5 calibration day(s) and the validation-day artifact its 4 validation day(s), none shared.
quality	both day sets	PASS	yes	Targets quality-masked: calibration days: runs/p1_rehearsal/dq/data_quality.json (sha256 8cae907dec40): 24 masked detector-day(s) (24 excluded whole), 912 reading(s) set aside; validation days: runs/p1_rehearsal/dq/data_quality.json (sha256 8cae907dec40): 19 masked detector-day(s) (19 excluded whole), 720 reading(s) set aside.
C1	calibration	FAIL	yes	Link flows, calibration days: GEH < 5 on 53.3 % of 840 station-hour comparisons pooled over 20 replicate(s) (per-replicate mean 53.3 %, 95 % interval 51.1 % to 55.6 %); the target is >= 85.0 %, short by 31.7 percentage points; hours anchored at the study period's start (1800 s).
C2	calibration	FAIL	no	Link flows, Texas criterion, calibration days: GEH < 3 on 39.0 % of 840 station-hour comparisons pooled over 20 replicate(s) (per-replicate mean 53.3 %, 95 % interval 51.1 % to 55.6 %); the target is >= 100.0 %, short by 61.0 percentage points; hours anchored at the study period's start (1800 s).
C3	calibration	FAIL	yes	Speeds, calibration days: RMSPE of 15-minute station mean speeds 39.7 % (mean over 20 replicate(s), 95 % interval 39.3 % to 40.1 %; point speeds at the stations, as a loop reads them); the target is at most 15.0 %, over by 24.7 percentage points (diagnostics: 5 min 41.0 %; 60 min 34.8 %).
C6	calibration	PASS	yes	Bottlenecks, calibration days: 0 observed bottleneck(s), 0 active for 30 min or more, compared with 20 replicate(s) (point speeds at the stations, as a loop reads them); all four rules hold.
C1	validation	FAIL	yes	Link flows, validation days: GEH < 5 on 48.8 % of 840 station-hour comparisons pooled over 20 replicate(s) (per-replicate mean 48.8 %, 95 % interval 46.1 % to 51.5 %); the target is >= 85.0 %, short by 36.2 percentage points; hours anchored at the study period's start (1800 s).

Check	Day set	Status	Gating	Statement
C2	validation	FAIL	no	Link flows, Texas criterion, validation days: GEH < 3 on 33.7 % of 840 station-hour comparisons pooled over 20 replicate(s) (per-replicate mean 48.8 %, 95 % interval 46.1 % to 51.5 %); the target is ≥ 100.0 %, short by 66.3 percentage points; hours anchored at the study period's start (1800 s).
C3	validation	FAIL	yes	Speeds, validation days: RMSPE of 15-minute station mean speeds 39.0 % (mean over 20 replicate(s), 95 % interval 38.7 % to 39.4 %; point speeds at the stations, as a loop reads them); the target is at most 15.0 %, over by 24.0 percentage points (diagnostics: 5 min 40.7 %; 60 min 33.1 %).
C6	validation	PASS	yes	Bottlenecks, validation days: 1 observed bottleneck(s), 0 active for 30 min or more, compared with 20 replicate(s) (point speeds at the stations, as a loop reads them); all four rules hold.
C4	calibration	FAIL	yes	Wave speed: a backward front in 20 of 20 replicate(s) (100.0 %; at least 80.0 % needed, a replicate without a front counting as a miss); simulated backward wave speed 5.4 km/h, 95 % interval 5.2 to 5.5 km/h over those replicates (stack detector); the band is 14-22 km/h, outside it by 8.6 km/h; observed on the calibration days (detector cross-correlation): median 19.1 km/h from 7 of 13 station pairs.
C5	all runs	PASS	yes	Collisions: none in 20 run(s).

Acceptance criteria

Criterion	Value	Threshold	Evaluated	Result
link_flows_geh	0.6774	GEH < 5 for > 85% of link-hour comparisons	yes	FAIL – fraction of comparisons with GEH < 5
wave_speed	5.375	backward wave speed in [14, 22] km/h (emergent, unseeded)	yes	FAIL – detector: stack: slant-stack peak of the two-way demeaned field on 15 s x 75 m bins over front speeds [-40, -2] km/h in 0.25 km/h steps, peak/median contrast ≥ 3 , edge peaks rejected
ring_emergence	–	Sugiyama ring emergence benchmark reproduced	no	NOT EVALUATED – not evaluated: input not supplied
ring_dampening	–	Stern single-AV dampening benchmark reproduced	no	NOT EVALUATED – not evaluated: input not supplied
n_seeds	20	n_seeds ≥ 20	yes	PASS
sensitivity_grid	–	penetration {1%, 2%, 5%, 10%, 15%, 20%} x compliance {25%, 50%, 80%, 100%} grid published with CIs	no	NOT EVALUATED – not evaluated: input not supplied
no_collisions	0	zero SUMO collisions in every run, each run recording the counter	yes	PASS – no collision in 20 run(s); FlowState internal standard (CLAUDE.md §3.3; owner decision 2026-10-04), not an FHWA or DOT criterion
no_locks	0	no run locked (no queue standing without discharge for 10 min or more), every run recorded	yes	PASS – no lock in 20 run(s); FlowState internal standard (CLAUDE.md §0.1; docs/I94_COLLAPSE_DIAGNOSIS.md §7, 2026-10-07), not an FHWA or DOT criterion

Criterion	Value	Threshold	Evaluated	Result
wlb_release_share (on-ramp 745524613)	0.01382	reported beside no_locks, not gating: Wlb releases as a share of the entrance's departures, pooled over the runs; above 1% flagged in Limitations	yes	REPORTED – reported, not gating: 281 of 20340 entrance departures released into the paired exit by Wlb over 20 run(s) (1.38 %), ABOVE the 1% design bound (flagged in Limitations); W2 over the runs it was on in: handback skips 3666, close-leader withholds 681, opposing deferrals 7018, of them vetoes 6946; Amendment 4 (docs/FRISCO_PROTOCOL.md, 2026-10-07): weave rules Wlb and W2 are part of the model; a FlowState disclosure, not an FHWA or DOT criterion
wlb_release_share (on-ramp 769818012)	6.522e-05	reported beside no_locks, not gating: Wlb releases as a share of the entrance's departures, pooled over the runs; above 1% flagged in Limitations	yes	REPORTED – reported, not gating: 6 of 91997 entrance departures released into the paired exit by Wlb over 20 run(s) (0.01 %), within the 1% design bound; W2 over the runs it was on in: handback skips 2924, close-leader withholds 93, opposing deferrals 16372, of them vetoes 15589; Amendment 4 (docs/FRISCO_PROTOCOL.md, 2026-10-07): weave rules Wlb and W2 are part of the model; a FlowState disclosure, not an FHWA or DOT criterion

The rows are the battery's, as it scored them. A row marked not evaluated or not recorded is never a pass, and an unevaluated criterion counts as failing; a row marked reported is a disclosure, never a pass or a fail.

Metrics

Mean with two-sided t-distribution confidence bounds at the 95 percent level over n replicates, as the battery computed them. Rows flagged underpowered have fewer than 20 replicates and must not be quoted as headline results.

Rows marked (model estimate) carry fuel (model estimate, SUMO HBEFA emission class; not validated against measured fuel). SUMO computes fuel from its HBEFA emission classes; this study has measured no fuel to check it against (docs/FRISCO_PROTOCOL.md section 8.6).

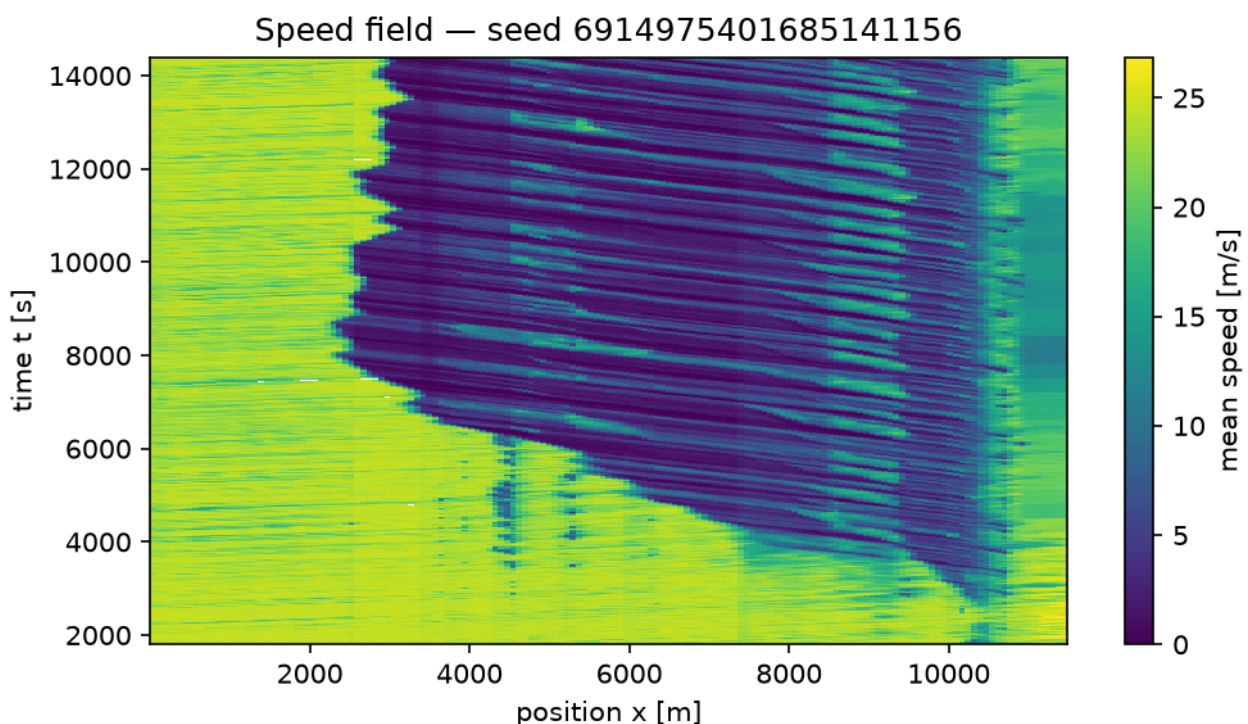
Metric	Mean	Lower	Upper	n	Underpowered
throughput_veh_h	2702	2692	2712	20	no
mean_tt_s	816.5	806.7	826.3	20	no
p90_tt_s	2095	2073	2118	20	no
sigma_v_spatial_ms	6.712	6.672	6.751	20	no
sigma_v_temporal_ms	4.839	4.809	4.869	20	no
vmt_veh_km	1.277e+05	1.274e+05	1.279e+05	20	no
vht_veh_h	5575	5499	5651	20	no
fuel_ml_per_veh_km (model estimate)	126.9	125.6	128.2	20	no
wave_count	28.8	26.63	30.97	20	no
wave_speed_kmh	7.333	6.972	7.694	20	no
wave_amplitude_ms	14.67	14.48	14.87	20	no
n_travel_time_veh	1.262e+04	1.258e+04	1.266e+04	20	no

Notes recorded with the battery

- Observed comparisons cover only the artifact's stations and windows; a station-window the detector did not measure is skipped, never imputed.
- The run's configured warm-up is discarded from every metric and from every observed comparison.
- The wave_speed row is measured with the profile's own detector on its own bins; validation.metrics.wave_speed_kmh in metrics_ci is the standard detector's separate diagnostic.
- The insertion block counts the vehicles the runs actually put on the network against their demand plan; a mean departed fraction well under 1 means the metrics above describe less demand than was configured.
- The locks block counts the seeds that locked (validation.locks: vehicles standing with no discharge past a point for the lock duration while a queue builds behind it); every metric, criterion and interval above includes those seeds, whose vehicles trapped behind the lock never finish.
- The weave_exits block counts, per weaving section, the exit-bound vehicles given up at the gore's end and rerouted through; the exit's link flow is short by missed_exit.n and every mainline link downstream carries them, so the geh rows are wrong by that count on those links (flagged above threshold_share of reached exiters).
- Ring rows not evaluated (ring-seeds 0); reported as failing per CLAUDE.md §0.1.

Speed contours

Figures from replicate 6914975401685141156, re-run with its trajectories by stage p22_reports; a contour covers the run's scored window (its warm-up dropped).



Space-time mean-speed contour, seed 6914975401685141156

Limitations

- At the on-ramp 745524613 weave, W1b sent 281 of 20340 entering vehicles (1.38 %, over 20 run(s)) into the paired exit after each stood the W1b dwell at the end of the auxiliary lane — above the 1 % design bound of Amendment 4. W1b is a safeguard that keeps the simulation running, not measured driver behaviour: a release ends either a lock at the gore or a long ordinary wait, the share counts interventions, not realism, and the exit's and the mainline's flows carry those vehicles.
- Model-integrity lines that need every replicate's run metadata beyond what the battery artifact

carries (forced lane changes, per-run wall times) are not repeated here; the battery's own auto-report carries them (docs/reports/mndot_i94_wb_stpaul_weave_xlsfg_dc_cal_w1b_w2_rbc/report.md).

- Criteria not passed by the battery: link_flows_geh, wave_speed, ring_emergence, ring_dampening, sensitivity_grid. The results are reported as they came out; no validation claim is made from rows that did not pass.
- The contours show one replicate (seed 6914975401685141156) of 20; the tables describe all of them. A single replicate's contour illustrates the configuration; it is not a statistic and not the replicate-mean field the speed criterion compares.
- Results cover a single corridor and scenario family; transfer to other corridors is not established.
- Model-form uncertainty (car-following and lane-change model assumptions, demand representation) is not captured by seed-to-seed confidence intervals.
- AV penetration and compliance are swept assumptions, not measured behaviour; no strategy result is part of this report.
- Every fuel figure is a model estimate (SUMO HBEFA emission class), not validated against measured fuel; differences in fuel between configurations are model predictions.